

chapter 2

RESEARCH FORMULATION AND LITERATURE REVIEW

Objectives:

After completing this chapter, you can understand the following:

- [The definition of problem and its formulation](#)
- [The review of literature](#)
- [The characteristics of a good research question](#)
- [The process of literature review](#)

Primary step in the process of research is selection of a research problem. The researcher needs to have a vast knowledge in the domain that he/she wish to work. Only then, he can analyze the gaps in the present situation and put forward a new proposal. For getting immense knowledge in their field, he/she should carry out vast literature survey. This chapter details how to select a research problem and the various aspects related to literature survey.

2.1 PROBLEM DEFINITION AND FORMULATION

The best explanation for research problem is provided by Northrop during 1966. He stated that “Inquiry starts only when something is unsatisfactory, when traditional beliefs are inadequate or in question, when the facts necessary to resolve one’s uncertainties are not known, when the likely relevant hypotheses are not even imagined. What one has at the beginning of inquiry is merely the problem”.

In simple words, we can say that research problem defines the destination before starting the journey. It specifies, what to do, how to do, where to do and what the outcomes are. Thus without a properly defined problem, the research cannot progress. Formulation of research problem is not just finding a topic within our interest. It is the remodelling, reshaping or even reconstruction of facts, theories or hypothesis. The problem thus formulated should also be compact for data collection and analysis. [Figure 2.1](#) shows the directions of problem formulation from a situation.

We can visualize this process like a magician’s magic box. Whenever we open the box, it contains another one. Similarly, the process continues and he takes something very precious from the innermost boxes. Likewise, in research we start with a big domain and we narrow it down to smaller pieces and finally gets to the core and obtains the real research problem.

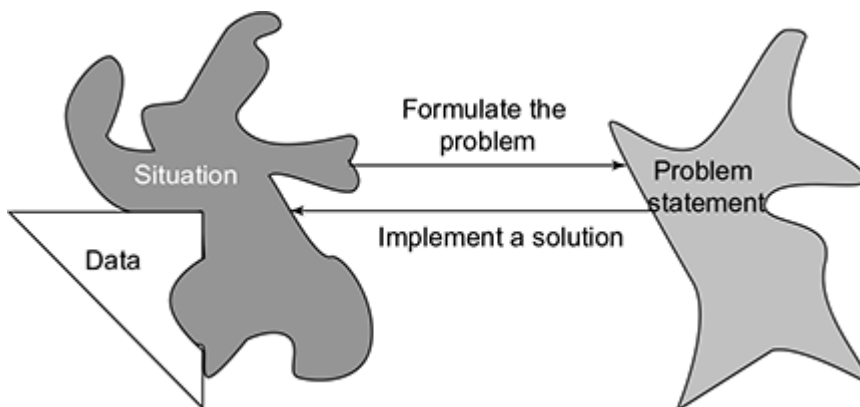


Fig. 2.1 *Research problem formulation*

The sole aim of problem definition is creation of research question and creation of hypothesis. So, what exactly is a research question? They are kind of questions that could be answered through the analysis of data. The data can be either qualitative or quantitative. It can also be the answers of some survey questions. Where hypotheses, on the other hand, are the guesses or conclusions generated from the theory that were created during data analysis.

In short, we can summarize the goal of formulating research problem as the method of generating measurable, well-defined, directed and in-scope research questions for creating desired hypothesis.

Case Study

Domain: Alcohol consumption among school children.

Theory formulation: Increase in price may reduce the demand

Research question: Will the price hike cause lower consumption of alcohol among school children?

Hypothesis: Increase in tax, reduces the consumption to a great extent

Here, a social survey is being conducted to study the use of alcohol among school children. One of the major causes of this is the easy availability of the product. The students have ample money with them to consume this. But what if, the prices are increased? Not all but still a particular percentage of students will not be able to constantly get the product. Thus, the consumption can be reduced, and a hypothesis is generated. In our social situation, the real question is, what will they do to get more money? A serious question to think about....

There are certain guidelines that need to be followed while formulating a research problem. The single statement that needs to be in every researchers mind is, there is no short cut to the research – it is only the hard work and determination that gives you the perfect result. Many scholars hastily skip the primary step of problem formulation, which makes them to face difficulties in later stage. The problem formulation cannot be borrowed; the guide can only throw light to the vast domain; it is the responsibility of each researcher to find the problem on his/her own interest. Having spotted the domain, right questions should be asked and answered. The non-biased and unattached approach will surely make way to have a core research problem. Formulation should start with a quick and dry study of the topic, and also need to have some alternative problems which can help in formulating our main scope. The problem thus formulated should be novel, significant and useful to practitioners.

Always keep the five **W's** in mind.

1. What is my research?
2. Why do I want to do this research?
3. Who are my research participants?
4. Where am I going to do the research?
5. When am I going to do the research?

Research problem identification is like treasure hunting. The hunting may have any shape (circle, square, rectangle, etc.) fields. These fields are research topic areas selected by a researcher. A shapeless field is considered as “no idea” in the research problem that has been selected by a researcher. He/She starts from outside of the field, slowly enters into the area and search the core (actual research problem). If he/she reached the core, then they will try to understand the core and try to move out to the surface. This journey is the problem solving. So there are two journeys – identification of the research problem by a wide search and how to solve the identified problem. During this travel, he/she may conduct a literature survey, reviews and becomes a scholar in that field. In this second travel, he/she has the research problem(s) and tries to identify how it can be solved. When they reach the surface, they win with appreciations (publications, patents, etc.)

[Figure 2.2](#) shows how a researcher identifies a research problem and solves this problem by methods. A basic confusion always arises is, where to start? Is any result will be obtained? Am I in the right track? etc.

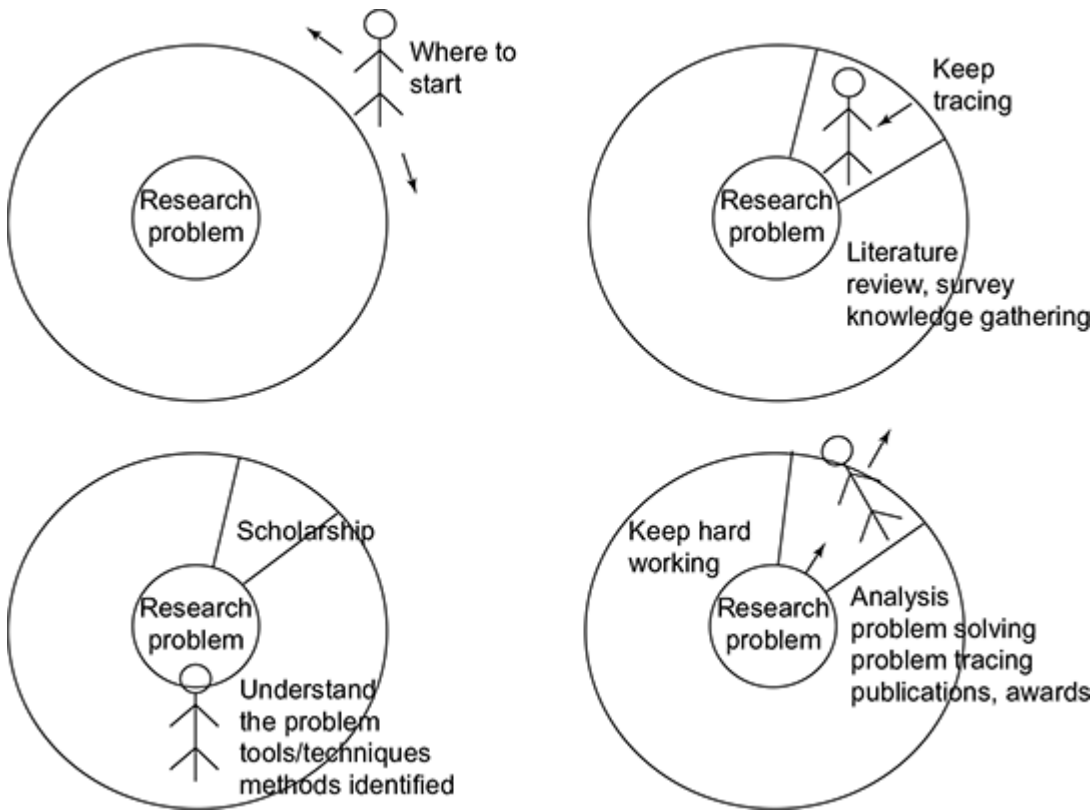


Fig. 2.2 *Identification of research problem*

2.1.1 Problem Selection

The research problem undertaken for study must be carefully selected. Major sources of research problem can either come from experience, observation, interest or needs. While choosing a problem, you should make sure that the area chosen is of real interest. Also an area with some practical value will help in a wide acceptance. A general feasibility study should also be done before a selection. [Figure 2.3](#) shows a sample feasibility study check-list. For almost all researches, pilot study adhere will be a sample check-list to evaluate the feasibility and work is carried out only when the feasibility score card is passed.

		YES	NO
1	Is the problem of current interest? Will the research result have social, educational or scientific value?		
2	Will it be possible to apply the results in practice?		
3	Does the research contribute to the science of education?		
4	Will the research opt new problems and lead to further research?		
5	Is the research problem important? Will you be proud of the result?		
6	Is there enough scope left within the area of research (field of research)?		
7	Can you find an answer to the problem through research? Will you be able to handle the research problem?		
8	Will it be practically possible to undertake the research?		
9	Will it be possible for another researcher to repeat the research?		
10	Is the research free of any ethical problems and limitations?		

Fig. 2.3 *Feasibility check-list*

One major criterion while selecting the problem is to avoid those areas that have already been overdone. For example, nowadays most sort of research activities in the area of computer science is dealing with cloud computing and its security. Even though selecting a problem from such an area helps you with a lot of literature, the probability of getting a very valid research problem is rare. Also subjects of sensitive scope or controversial topics should not become the choice unless you are ready to face the consequences. Those vague and broad areas that narrow our working space should also be avoided from selecting. [Figure 2.4](#) gives how research problem is evolved.

Another major factor that needs to be considered is the availability of sources that perform a particular research. For example, if we do not have any particular arrangements and sources to study the presence of water in the planet Pluto, well then the research will just end in nowhere. So availability of resources should be checked before selecting a problem. Before selecting, some experts and universities that are working in your area should be contacted. This can help you a lot in future.

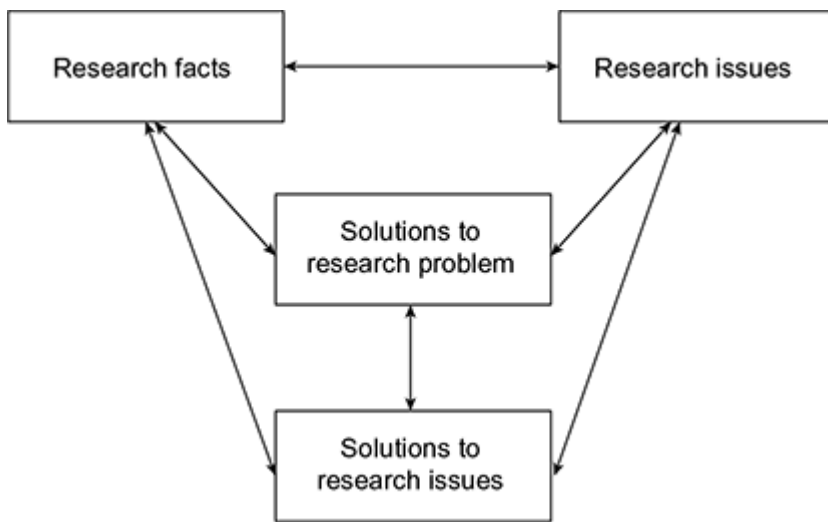


Fig. 2.4 *Research problem evolution*

The cost and time involved in the research should also be a key point while selecting the problem. Even conducting a pilot study of the problem, a researcher must know the present situations, drawbacks and the current state holds. Reading latest articles can also help him/her to achieve these goals. The genuineness of the research problem also needs to be checked. Anticipate the results, critically question, discuss with the colleagues and finally focus on the goal. All these help you to get the diamond (research problem) out of the shell (domain).

If the research problem is selected appropriately, by conforming to the aforementioned points, then the research will not be a boring drudgery. Rather, it is exciting and educating. The selected subject/problem must involve the researcher and be the prime priority in his/her mind, so that he/she may give his/her best shot required for the study.

2.1.2 Necessity of Problem Definition

“A problem well-defined is a problem half-solved” holds strong even today. Proper definition of a research problem is a prerequisite for any research study. Formulation of a problem is more significance than its solution. The entire direction in which research moves is decided by the way in which the problem is defined. The definition will help in differentiating the relevant and irrelevant data. The process model, design of solution, steps that are involved in solving the research problem can easily be spotted if the problem is well defined.

- What data needs to be collected?
- What characteristics of that data are relevant and need to be studied?
- What relations have to be explored?
- What techniques have to be used for the purpose?

All these questions can easily be answered and if the problem is well defined. Overall quality of the research and efficiency will increase only when the system is well defined. Problems which are ill defined can cause hurdles in the path of research. The techniques that need to be incorporated in our study can be initialized and selected if the problem statement is clear. In short, we can say that a research problem should be FINER (Feasible, Interesting, Novel, Ethical, Relevant). A well-defined research problem keeps the researcher in track and helps to focus his/her work. [Figure 2.5](#) gives steps involved in research problem making by a researcher.

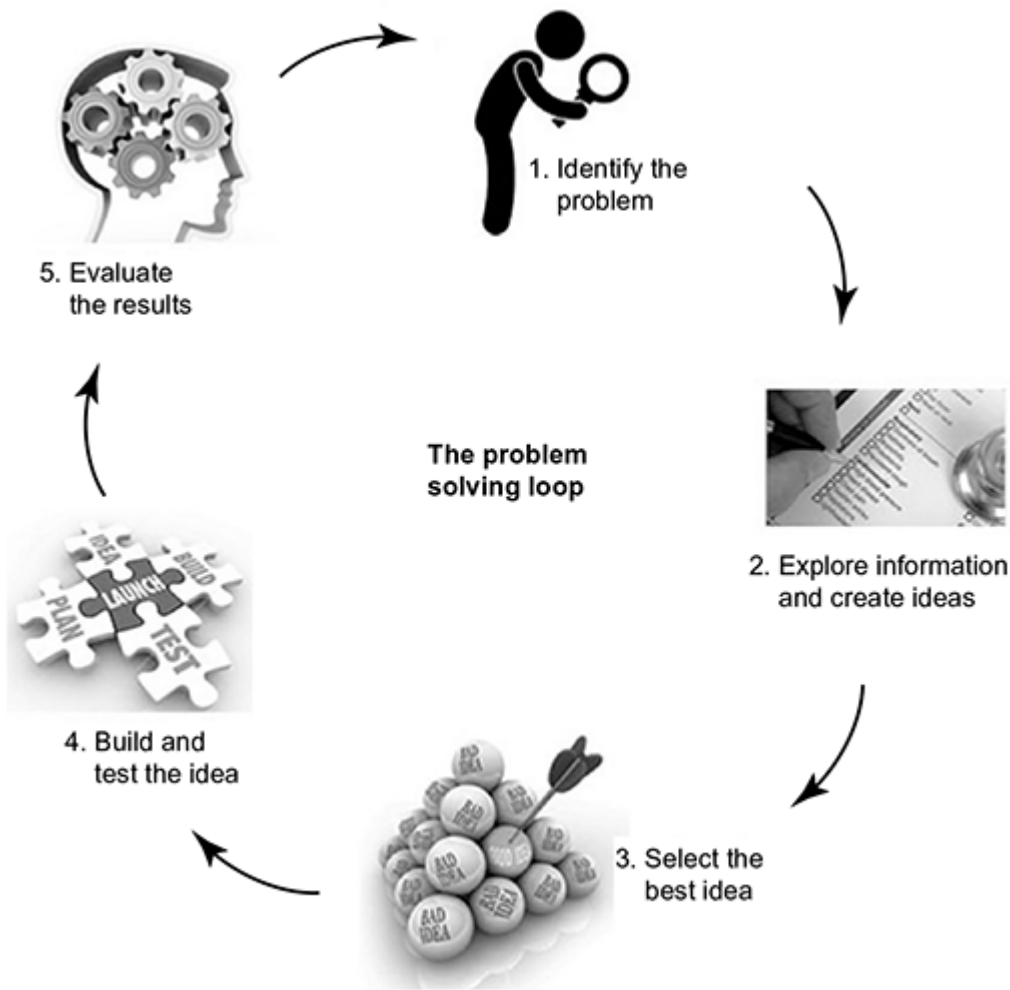


Fig. 2.5 *Research problem loop*

Problem formulation

This section illustrates an example on how to formulate a research problem. Consider a general problem,

Why the productivity in USA greater than in India?

While working in this area, there are many questions that need to be resolved. Starting from, what kind of productivity is the domain talking about? With what industry does the comparison takes place? What period of time does this comparison valid? While answering such questions the general domain becomes much specific. Now we can modify our question to, why productivity of America greater than India, with regard to IT industry from the time period 2005–2008. We can further rethink and reorganize our questions like, what are the major factors? What amount of increase does happen and likewise?

2.2 LITERATURE REVIEW

Literature means writings, and a body of literature refers to all the published writings in a particular style on a particular subject. In research, a body of literature is a collection of published information and data relevant to a research question. Literature survey is not a compilation of every work written about a particular topic. It is a survey or overview of the literatures found to be significant to the area under study. Review of literature always help you to increase your knowledge of your topic, identify important authors and works in your area of research, identify new researches, theories and/or methodologies in your area of research. While doing literature review, it should not stand as a stand-alone document. It should include an introduction defining your topic and the purpose of your review of the literature. It needs to be organized by common themes or categories. It should

also contain the summary and analysis of each work including its importance to the overall topic as well as its relationship to the other referred works.

2.2.1 Critical Analysis

A critique is a person who evaluates somebody's work (e.g., book, essay, movie, painting, article, etc.) in order to increase its understanding. Critical analysis is a view point expression of a writer who evaluates the work. So it is a subjective text to break down and study the parts. Critical reading and critical writing are the two parts of critical analysis. Critical writing depends on critical reading. An involvement is reflected in the texts of a researcher's written text who is already interested. The interpretations are like a judgement and formulate findings based on your approach. In critical analysis, the following type of questions may be considered by a researcher.

Theoretical questions

Questions are prepared based on a theory.

For example,

- How authors are concluding their views in this situation?
- What is the relevance of sigma value?

Definitional questions

Questions are prepared based on definitions.

For example,

- Does the author consider two different aspects?
- Is any relevance in comparing communism and democracy in China?

Evidence questions

Questions are prepared based on selected evidence.

For example,

- Do authors give evident support for their arguments?
- Do regulatory mechanisms are applicable to this gene?

2.2.2 Critical Thinking

Critical thinking, according to National Council for Excellence in Critical Thinking, is the intellectually disciplined process of actively and skilfully conceptualizing, applying, analyzing and/or evaluating information gathered from, or generated by, observation, experience, reflection, reasoning or communication, as a guide to belief and action. One of the key questions of research is how do you infer into a correct decision from you problem? Thinking is one of the basic solutions of the problem formulating skill. Is all thinking formulated into correct problems? If your approach as a critic to your problem may give much idea into your research problems, there come certain skills in your thinking. [Figure 2.6](#) explains the required skill sets of a researcher.

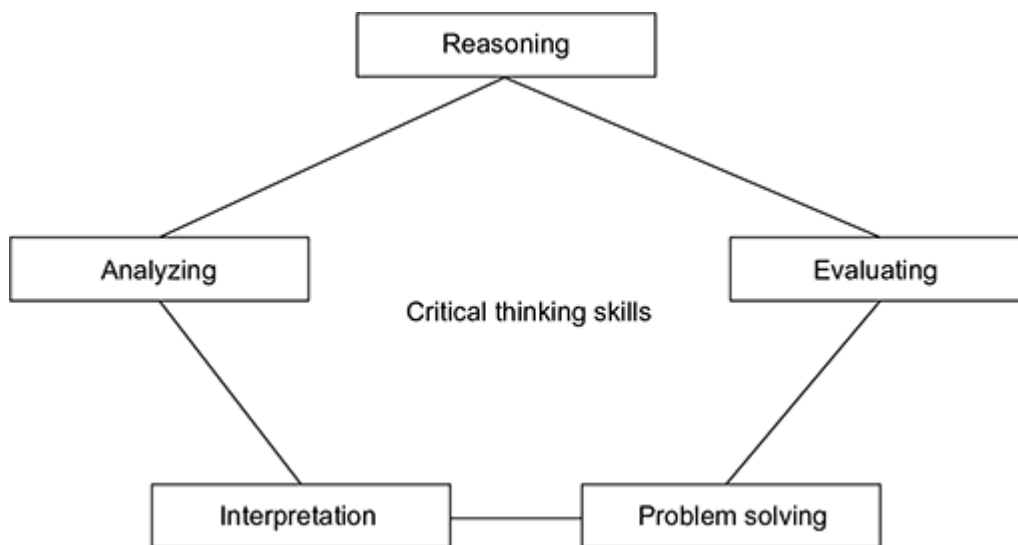


Fig. 2.6 *Critical thinking*

Interpretation is the ability to understand the information you are being presented with and being able to communicate the meaning of that information to others. During the analyzing step, a critic separate or break a combined idea into parts to notice their nature, functional and relationships. Reasoning is ability to understand and recognize what elements you will need in order to determine an accurate conclusion or hypothesis from the information you have at your disposal. At the evaluation stage, the identified statements are judged or meet an opinion to measure the validity of the presented information. Problem solving is a research skill that arrived at a set of conclusions from a question.

2.2.3 Critical Evaluation

A critical evaluation of a work must consider provenance, objectivity, persuasiveness and value. Provenance shows author's credentials or author's arguments supported by evidence.

For example, a researcher reveals his ideas of historical materials with some case studies or researcher needs to publish his recent scientific findings. Objectivity is the author's perspective of even-handed or prejudicial mind in the secondary data analysis. For example, a researcher discusses "cause and effects of atomic energy", but significantly ignored the first atom bomb blast. Persuasiveness is author's convincing points to a narration of subject. For example, a researcher likes to explain growth of a new political party in middle-class people. The author remarked some convincing points such as inflation or scam produced by other parties, which are major reasons of new party growth. Value is the author's convincing argument and conclusions. For example, researcher gives ultimate contribution of his/her understanding of the subject that he/she had discussed.

2.2.4 Objectives of Literature Review

Authors should try to accomplish the following four important objectives in preparing a literature review:

1. The review should provide a thorough overview of previous research on the topic. This should be a helpful review for readers who are already familiar with the topic and an essential background for readers who are new to the topic. The review should provide a clear sense about how the author's current research fits into the broader, sociological understanding of the topic. When the reader completes reading of the literature review, he/she should be able to say, I now know what previous research has learned about this topic.
2. The review should contain references to important previous studies related to the research question that are found in high-quality sources such as scholarly books and journals. A good literature review conveys the readers that the author has been conscientious in examining previous research. Authors in the research

build on what is already known. In this process, highly interested readers are also provided with a set of references that they may wish to read themselves.

3. The review should be succinct and well organized. Most scholarly journals stipulate a maximum length for papers submitted for publication, and often this is only about 20 pages. After all the work that has gone into a paper, authors typically feel that they could write at least twice that much. Thus, every page is precious, and authors must learn how to write succinctly. A typical literature review is only about 3, 4 or 5 of the 20 pages, and it must contain a lot of information. Therefore, it is necessary and helpful to learn how to do it succinctly.

Many authors like to begin with a short “Introduction” section that identifies the general topic and its importance. This is followed by the “Literature Review” section that provides the overview of the previous research and explains what has and what has not already been learned. Much of the focus of literature review is on previous research related to the dependent variable. This includes use of sociological theories to explain the dependent variable. It may also be appropriate to focus on research related to the independent variables.

4. The review should follow generally established stylistic guidelines. This conveys to readers that the author is familiar with scholarly publication style and that can add legitimacy to the author’s work. In addition, when the typical style is used, it is easier for readers to immediately follow the paper’s organization.

2.2.5 Importance of Literature Review

“Literature review is a scholarship”

How will you improve your knowledge before conducting an experimental or theoretical work? The available literature should be reviewed related to your study and requirements. Literature review is a survey of scholarly articles, books and other sources relevant to a particular area of research, issue and theory. Output of the literature review is a description, summary and critical evaluation of selected works. For example, a researcher wishes to develop an expert system for power distribution substations. He/She should review most of the works related to power plant and power distribution expert systems before his/her actual work design. Without proper literature review, the researcher will not be aware about similar works done in that area. It will end up in creating work plagiarism.

Researcher becomes a scholar in his/her area by a proper review in the literature. A literature review offers simple summary of the key sources with organizational patterns and results. The following are the purposes of the literature reviews:

- Study the contribution of others’ work and its contribution of the research work in the problem domain.
- Identify the relationship of each work to others and note them before the actual work.
- Identify the gaps in the previous research by new interpretations and drop light on any identified gap.
- Analyze the conflicts and resolve it, if any in the previous studies.
- Duplication efforts can be prevented by prior scholarship in the current field.
- Additional research area can fulfil after a successful literature review.
- Literature review reveals common findings among studies and discloses inconsistencies between the studies.
- Literature review identifies factors not previously considered and provides suggestions for further research.

Literature review is important because

- Output of the literature review describes how proposed research work is discriminated from the previous research statistics,
- Originality of the content and relevance of the proposed research problem,
- Proposed research methodology is justified,

- Literature survey pre-demonstrate the research proposal.

Case Study

A researcher wishes to develop a language translator from a scrambled English text. He/She would develop a system that identifies Malayalam words from a SMS or e-mail message. The message is typed in English letters. How he/she can identify the Malayalam words typed in English letters? Before entering to the work design, he/she should review all the literatures that give related study. Available literature may give ideas in English to Arabic translator or English to Hindi translator. That is, the words typed in English language letters for Arabic or Hindi words. These literature reviews accelerate his/her research work and produce better results. He/She can write a survey article from the above studies.

2.3 CHARACTERISTICS OF A GOOD RESEARCH QUESTION

Certain characteristics are required in the creation of good questions in research.

1. A good research question can be answered by collecting and analyzing data.

Your literature search will be limited to scholarly research, which nearly always attempts to answer questions by gathering and analyzing the data. If you ask a question that cannot be addressed by data, you will not have a research question. An example of a non-research question would be one like this:

Will text messaging be the end of spelling as we know it?

That is an interesting question, but inappropriate as a research question because data could not answer a question about behaviour in future.

Another non-research question is:

Do parents have a moral obligation to be involved in their children's schools?

Although, worthwhile for individual consideration, questions of morals, values and religious faith cannot be answered by data collection and analysis. However, you could survey a group of people to determine if they believe that parents have a moral obligation to be involved in their children's education, but you would then be asking a different question.

2. A good research question assumes the possibility of different outcomes or opinions.

What is the Individuals with Disabilities Education Act? There is no room for opinion in this question. Any credible source will provide identical information (though some sources will provide more and some less). There is no way to draw a conclusion from it. This question might be the start of a report, but not of a literature review or research design, which requires presentation of various points of view in order to add new insights.

How did No Child Left Behind affect the main streaming of students receiving special education services?

This is a better research question because it will like generate more than one viewpoint, that allows the writer to reach a conclusion (positive impact, negative impact, no impact, various impacts, etc.). The question can also be addressed by the collection and analysis of data, and it is limited in scope main streaming, special education.

3. A good research question is narrow.

What makes a good teacher? Think about what kind of research could answer this question. First of all, what does "good" mean? How could "good" be measured or defined? Second, who should answer the

question: Teachers, Students, Parents or Administrators? Second, would the answers be the same for each grade level? Third, what kind of teacher is to be included: science teacher, Tutor, Band director, Economics professor or Parent? You have got the picture. The question is too broad.

4. A good research question is clear.

What is the best way to teach sex education? Such a question like this would certainly involve a diversity of viewpoints; however, it is also too broad and vague to be meaningful. There is the problem of deciding what “best” means and how it would be measured. There is a question about who would be doing the teaching (public school, parents, religious institution, etc.) There is the problem of context: age and gender of children, type of school and so on.

One better way to ask a sex education question is:

Can public middle school sex education classes significantly reduce student pregnancy rates?

This is a better question because it is limited (public middle school, “good” = significant reductions in student pregnancy rates). Although this is a yes/no question, you are likely to uncover a pattern of successful sex education; in other words, what kind of instruction resulted in the greatest reductions. In that case, you will have a rich field to harvest for your discussion and conclusion sections of the literature review.

5. A good research question is a single question.

Sometimes research questions have corollaries (closely related sub-questions), but for now just focus on a single guiding question for which you attempt to find an answer in the existing literature. The single guiding question should not have a multi-part. For example, do not ask, how larger class sizes affected student test scores, student behaviour and teacher job satisfaction? Obviously, these are three questions, and not one (in addition to being too broad and vague). You will not be able to manage a paper length of a literature review, if multiple questions are embedded in your research question.

How do teachers and students benefit from teacher in-service training? That is two questions requiring two literature reviews.

6. A good research question is built on sound assumptions.

To save money, I am going to buy a hybrid car. Where can I find the cheapest hybrid? The underlying assumption is that buying a hybrid car will save you money. If so, finding the cheapest hybrid would be worth the search. But will buying a hybrid save you money? Independent consumer research shows probably not, depending upon what you are driving now, how much you drive, and the price of fuel. If saving money is the goal, then why research hybrid prices instead of researching cheaper cars that would save you more money?

How can schools get parents involved in their children’s education?

The question contains a strong assumption that children would benefit from their parents’ getting involved. Once you have clarified the meaning of “involved in their children’s education”, you must make sure that the benefits of such involvement have been supported by research and that the drawbacks of such involvement do not overshadow the benefits. If your assumption is faulty, you have lost your readers at the first paragraph.

Check your assumptions before you decide on the question. The validity of the assumption need not be established in the paper, but most readers will know if it is faulty or weak.

Most of the primary research study begins with a review of literature. Literature review offers a big picture of topic of research from the previous studies to a current researcher in that field. The literature review serves a path for additional research in the present problems by explaining the topic of research. Literature review results in a critical thinking in the summarized articles of present topic. Major steps involved in the literature review are given in [Figure 2.7](#)

1. Research topic identification

Students may choose research topics that are conceptualized and sometimes not reachable. They start reading the available materials and investigate their scope in the research problem. For a narrow defined problem, it may not be possible to identify any prior research that addressed to precise topic. A variety of sources can be selected with respect to researcher's interest, knowledge of social conditions, observations, challenges and specific topic. This may lead the researcher to primary research topic. Researcher can conclude a comprehensive review of the literature as his/her own for later use.

2. Review secondary sources to get an overview

To gather overview of the research, identify the secondary sources of the topic which is important. Many literature reviews are available as a secondary source. Someone may already write a good literature review in your research topic and is published. This may be useful for further reading. Journals, conferences proceedings, research communities, magazines, etc. are such resources for a researcher.

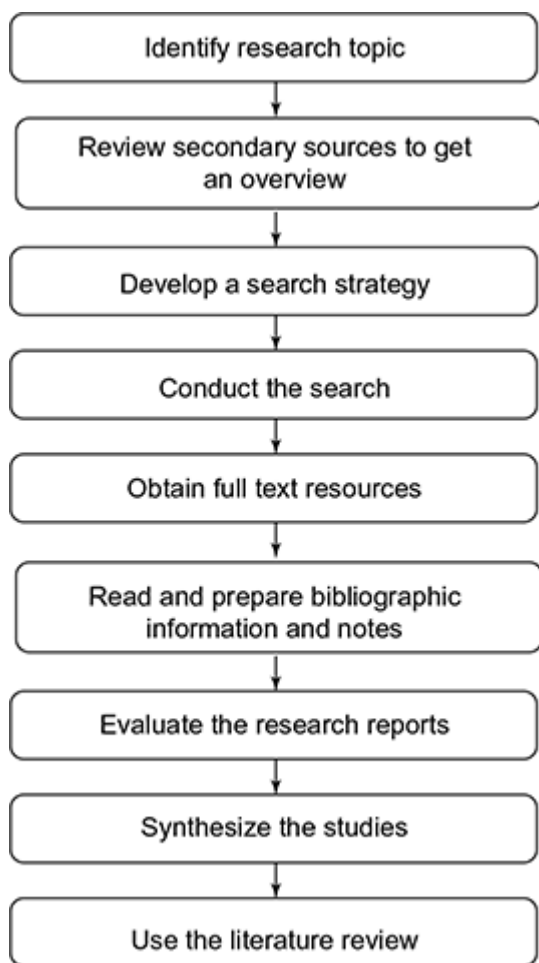


Fig. 2.7 *Process of literature review*

3. Develop a search strategy

Using the appropriate primary resources such as journals, conference proceedings, community networks, etc., researcher can develop a search strategy. Four important search strategies are identify preliminary sources, identify primary research journals, access personal networks and involve community members. Databases are preliminary sources that contain compiled indexes of bibliographic information, abstracts and sometimes full-text articles. These resources can preserve in a wide range of topics and are accessible in print form, in electronic media (CDROM/DVD) or in Internet. To identify primary research articles, first examine the reference list at the end of the relevant journal articles or books. Then directly go to the journals that is related to your topic of interest. For example, “Nucleic Acids Research” is a journal for researchers in bioinformatics or molecular chemistry, while “nature magazine” gives a wide range of research readers.

4. Perform the search process in the selected articles to review

Combination of self-knowledge and community knowledge and skills result in an effective research and evaluation methods. For example, research guides can organize their peer group internal meetings among the similar topics of interest. In the meeting, the researcher will be able to communicate with each other and discuss their topics. In such internal meeting, the researcher may get feedback from his peer group members, before communicating a work into a conference/journal.

5. Collect the full-text references

Many journals, articles, conference proceedings and books are available in online texts. The materials that are not available online can be accessed through the library. Librarians can provide articles by the help of complete bibliographic information. Most libraries are providing digital facilities to their members.

6. Prepare bibliographic notes from the articles and notes on each article

Read the collected documents and check whether it is required in your research. Relevant materials related to your topics need to be recorded as bibliographic information. These materials can be sorted and kept for further references. Several options are available for recoding the collected materials. One of the citation options is as follows:

Journal:

Author’s Last Name, Initials. (year). “Title of journal article”, *Title of Journal*, **volume number(issue number)**, page numbers.

For example:

Chandra V. (2010). “MTar: a computational microRNA target prediction architecture for human transcriptome”, *BMC Bioinformatics*, **11(S2)**, 1–9.

Book:

Author’s Last Name, Initials. (year). “Title of book”, **edition**, *Publisher*, Place of publication.

For example:

Chandra, V; Hareendran, A. (2014). “Artificial Intelligence and Machine Learning”, **1 edn**, *Prentice Hall*, India.

7. Evaluate research reports and make a note

Evaluation of reports is one of the special tasks of a researcher by making sufficient notes and backlog references. These reports are useful for further reading.

8. Identify and record your findings

Recording and storing the material is bookkeeping in the research work. Develop a flexible framework that synthesis stage by stage. It is flexible because you can add, delete and modify your organized materials. This is done by continuous review process. For example, in the second edition of a book, the author may have different organization style used and is not even mentioned in the first edition of the same book. After publishing the first edition, the author records these findings.

9. Use the literature review

Researcher must formulate a conceptual framework and plan research questions, hypotheses or both. A researcher formulates a conceptual framework that influences a high-level planning and conduct the research review. The researchers keep an open mind for self-arguments and question making. He/She can find hypothesis for his/her study and test whether they are correct.

A literature review is structured by an overview of the subject, issue or theory and objectives of the study. The division of literature review is organized into categories or subsections. Literature review focusses to an explanation of the similarity and difference with others' work. The written article needs conclusions that contribute your understanding and development of the area of research. The reviewed literatures are useful for later reference.

2.4.1 Primary and Secondary Sources of Literature

Primary sources are the first hand evidence left behind by participants or observers at the time of events. They are from the time period involved and have not been filtered through interpretation or evaluation. They are original materials on which other research is deployed. They are usually the first formal appearance of results in physical, print or electronic format. They present original thinking, report a discovery or share new information. Artefacts (e.g., coins, plant specimens, fossils, furniture, tools, clothing, all from the time under study), Audio recordings (e.g., radio programs), Diaries, Internet communications on email, Interviews (e.g., oral histories, telephone, e-mail), Journal articles published in peer-reviewed publications, Newspaper articles written at the time, Original documents, etc. come under primary sources of literature.

Secondary sources are less easily defined than primary sources. Generally, they are accounts written after the fact with the benefit of hindsight. They are interpretations and evaluations of primary sources. Secondary sources are not evidence, but rather commentary on and discussion of evidence. Bibliographies (also considered tertiary), Biographical works, Commentaries, Criticisms, Dictionaries, Encyclopaedias, Histories, etc. are the examples of secondary literature. [Figure 2.8](#) shows primary and secondary literatures.

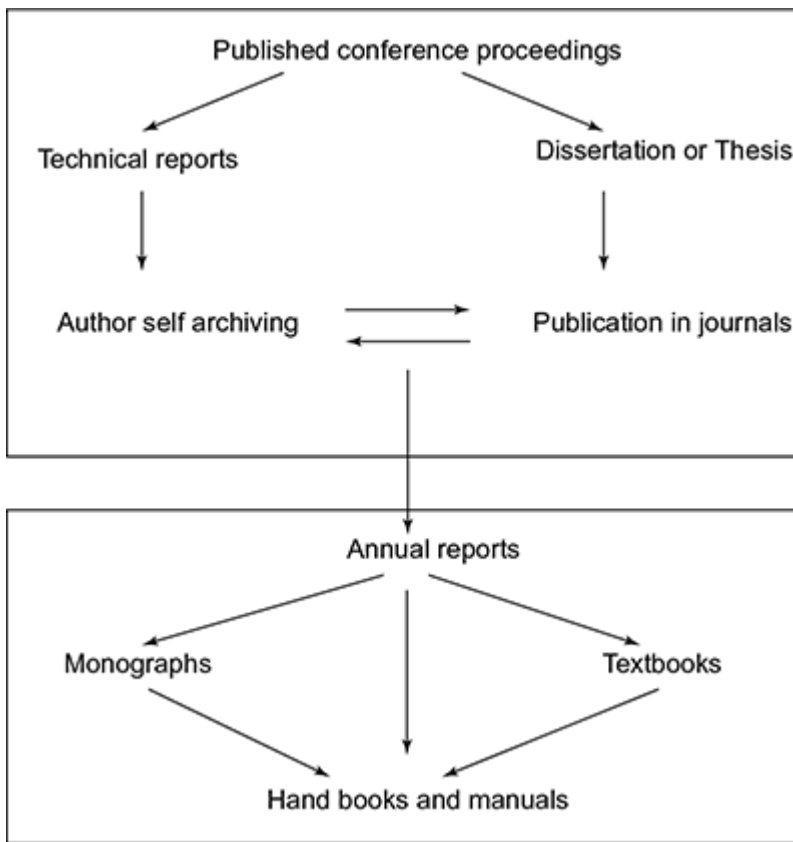


Fig. 2.8 *Primary and secondary literatures*

2.4.2 Identifying Literature Review Gaps

A gap in the literature is a research question relevant to a given domain that has not been answered at all or adequately in existing peer-reviewed scholarship. A gap in the literature may emerge if,

- The question has not been addressed in a given domain, although it may have been answered in a similar or related area.
- The question has never been asked before, but it now merits exploration due to changes in accepted theory, data collection technology or culture.
- The question has been asked and tested in peer-reviewed research, but the methods were either of questionable validity or had necessitated limited applicability of results. Alternatively, a replication study could be run to verify a published study's results if appropriate.

2.4.3 Literature Review Pitfalls

Is Internet a good literature resource? We can say that the Internet provides huge amount of data and information. Its service is countless for a researcher. But we cannot fully trust the Internet because the ownership of the data and information shared in Internet are not trustworthy. The Internet provides you pseudo-science and poor research. The documentations provided in the Internet are not correct and may be wrong path for a researcher. Therefore, identification of correct materials is one of the challenges of the researcher.

Another pitfall in the literature review is trade magazines and journals. Without much reviews or evaluation, some articles published or recommended for publication. These articles may not give any results or they may give some false results for the followers. These may misled a researcher, if he/she selects that article as a basic material.

In certain cases, students may not have right of entry to certain information. They have spent unnecessary time and resources on searching for the review. If the supervisor is not providing a right direction and feedback, the student may enter in to a confusing stage or their valuable time may be lost.

The good and poor literature reviews are obtained by researchers' views and hard work. [Table 2.1](#) shows a comparison of good and poor literature review output.

Table 2.1 *Comparison of good and poor literature review*

Good literature review	Poor literature review
It is a synthesis of available research	Looks like an annotated bibliography
It is a critical evaluation	That confined to description
It has appropriate breadth and depth	Review narrow and shallow
It has clarity and conciseness	Review text confusing and long winded
It uses rigorous and consistent methods	It is constructed in an arbitrary way

EXERCISES

1. Explain scientific importance of primary and secondary sources of literatures.
2. Can we trust Internet as a primary source of literature?
3. What is a literature review? How it differs from literature survey?
4. How do you arrive that the literature review is poor?
5. Explain the steps involved in formulating a research problem.
6. What are the major criteria for selecting a research problem?
7. Explain feasibility of a research problem.
8. Why is it so important to have a clear/specific research problem definition?
9. Explain critical thinking?
10. What are the major objectives of literature review?
11. What are the basic purposes of having literature survey?
12. Explain in detail, the various characteristics of a good research question?
13. Explain the literature review process.
14. Define and differentiate good and poor literature survey.
15. What do you mean by the statement, a good research question is a single question?